Mikołaj Płachta

Machine Learning Engineer, PhD Candidate

and experienced iOS & Android Developer

+48 793029135 linkedin.com/in/mxn-dev

mixxenxd@wp.pl Warsaw, Poland

JOBS

Reformat Studio sp. z o.o.

AI/Machine Learning Engineer
JUNE 2025 – AT PRESENT

Preparation and development of machine learning-based models for generating various types of raster graphics from supplied vector graphics. The solution is designed for operations on large data sets and: - analysis of available machine learning-based solutions for generating raster graphics
- preparation of a complete ELT process for feature extraction from psd/png files and preparation of a data warehouse based on Apache Spark
- research on the subject and preparation of various neural network models to perform this task
- use of transfer learning methods to accelerate the learning process
- orchestration of MLOps processes using AirFlow
- use of AWS cloud solutions for data processing (S3, EMR, MWAA)
Used technologies: **Python, PyTorch, Pillow, NumPy, PySpark, AWS(S3, EMR, WMAA), AirFlow**

Secure Intelligent Methods for Advanced Recognition of Malware and Stegomalware - SIMARGL

Machine Learning Engineer & Researcher
JULY 2020 – AT PRESENT

Participation in a research project seeking detection methods for image steganography. The work on the project was linked to the research of a doctoral thesis, the purpose of which is to seek effective methods for detecting Internet threats using deep machine learning methods. The main responsibilities were:
- literature analysis of available image steganography detection techniques
- preparing the data used for research using steganographic algorithms into matrices from Matlab
- developing of neural network models in different configurations of the number of layers, optimizers and activation functions using the Keras and PyTorch library
- testing convolution layers and replacing them with advanced feature extractors like GFR, DCTR and PHARM
- configuration of a test environment based on Ubuntu along with a data warehouse based on Apache Spark
- preparation of a tool demonstrating the performance of the prepared neural network model in steganography detection
Used technologies: **Python, Keras + Tensorflow, PyTorch, scikit-learn, SciPy, NumPy, matplotlib, Spark, Matlab, C++**

SpaceOS Ltd

Lead/Senior iOS & Android Developer
MARCH 2018 - JUNE 2025

Development of mobile applications for managing coworking spaces (resource reservation, news, events, user support, community, access control, etc.) for iOS and Android platforms using the principles of Reactive Functional Programming, Dependency Injection and MVVM depending on the platform and:
- work on rewriting iOS applications from Objective-C to Swift along with rewriting Android Java to Kotlin language
- implementation of a new UI with custom controls
- automation of branding and deploying applications to test channels using CI/CD processes (Jenkins/fastlane) along with full automation of distribution in mobile stores
- integration with external service providers using their SDK or direct implementation of BLE/NFC solutions for building access control
- writing unit tests and support in integration testing
- preparing documentation for specific modules of both mobile apps
- managing the entire mobile team in terms of setting development goals, recruitment process and overall mobile application development in terms of technology
Used technologies: **Objective-C, Swift, RxSwift, UIKit & Storyboard, Java, Kotlin, RxJava, Swinject, Dagger, Cocoapods, Fastlane, Jenkins, Firebase, BLE, NFC, Xcode, Android Studio**

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Inteliclinic sp. z o.o.

iOS & Android Developer
JULY 2017 – FEBRUARY 2018

Creation and development of mobile applications for iOS and Android platforms that integrate the functionality of the sleep analysis mask with the user's phone. Applications were written in accordance with the rules of RFP, using patterns such as DI or MVVM. The applications were also natively integrated with a low-level library written in C++ via Swift and JNI. In addition to development work, support for internal tests, distribution of applications through the Fabric service for internal tests and to the store for general distribution. Three applications were developed:
- Neuroon Open iOS - application for users of a sleep analysis mask, with modules for viewing extensive sleep statistics, profile management and creating alarms
- Neuroon Research iOS - application for mask testers to collect data from sleep, having two modes of operation: preparation of the test program and conducting a study with data collection, an application designed to support two different view modes for iPhone and iPad
- Neuroon Research Android - application for mask testers only for collect signals from this device
Used technologies: **Swift, RxSwift, UIKit & Storyboard, CoreData, RxBluetoothKit, Kotlin, RxKotlin&RxJava2, Swinject, Kodein, Cocoapods, SnapKit, Lottie, BLE, Xcode, Android Studio**

Execon ITS sp. z o.o.

iOS & Android Developer
AUGUST 2016 – SEPTEMBER 2017

Development of mobile applications for iOS and Android platforms, support with internal and external tests at the client's site, contact with the client to determine the scope of the project and planned work, partial project management (separation of mobile/server-API tasks, etc.), requirements analysis and application distribution to Google Play/App Store, working with applications:
- Gaspol Mobile - customer portal application for individual users, developing completely new version of the application for iOS and Android platforms and subsequent minor rebranding changes
- Gaspol Butle - application for individual users used to search for the nearest cylinder exchange points, implementation of new functionalities and creating a completely new rebranded version of the application
- internal GRM Mobile project - customer portal for the GRM project on the Android platform
Used technologies: **Objective-C, UIKit & Storyboard, CoreData, Java, GSON, ButterKnife, Google Play GCM, Maven/Gradle, Push Notification, CoreGraphic, CoreAnimation**

Nationale-Nederlanden U.F. S.A

Cobol Developer
JULY 2015 – DECEMBER 2015

Working in Cobol and IBM DB/2 databases on the Mainframe AS/400 system, developing new functionalities, fixing errors and support in tests.

Private order for the company PIW EKSPRO

Web Developer
AUGUST 2015 – NOVEMBER 2015

Custom website design for an external company, written in PHP and HTML using the MySQL database, website address http://www.ekspro.pl

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EDUCATION

Warsaw University of Technology – Doctoral School

Doctoral School
FEBRUARY 2020 – AT PRESENT

Field **Technical Computer Science and Telecommunication** research title: **Malware detection using deep neural networks**
The first part of the research for the dissertation was performed mainly within the SIMARGL project. The second part of the research concerns the analysis of heat maps created from the transformation of network traffic with potential hidden data. The same technologies as in the SIMARGL project are used. Mainly neural networks with convolution layers and basic neural networks with adapted feature extractors are tested. The research work is in the final stages of completion.
During the research work so far, 4 articles have been written:
- Detection of Image Steganography Using Deep Learning and Ensemble Classifiers (2022)
- Detekcja steganografii obrazowej z użyciem prostych sieci neuronowej (2022)
- A Simple Neural Network for Detection of Various Image Steganography Methods (2022)
- Comparison of computational complexity of neural network models for detection of steganographic algorithms (2024)

Warsaw University of Technology – Faculty of Electronics and Informational Techniques

Intramural master's studies
FEBRUARY 2016 – MARCH 2019

Field **Computer Science** specialization Information Systems Engineering (title of Master Thesis: **Mobile recommendation system using data from social networks, Markov chains and current history of user's visited places**)

Warsaw University of Technology – Faculty of Mechatronics

Intramural engineering studies
OCTOBER 2012 – FEBRUARY 2016

Field **Automation and Robotics** specialization Industrial Computer Science (title of Engineer Thesis: **iOS application for control marching robot**)

PROJECTS

iOS + Android applications
SEPTEMBER 2017 – AUGUST 2021

Locusity

Applications for iOS and Android platforms for mobile users as part of the master's thesis, used to recommend places to visit based on the history of places visited by the user. The application extensively uses data about the location, pedometer and barometer to predict the visited buildings. Interface designed with emphasis on good UI/UX and MVVM architecture.

iOS *Android*

Used technologies: **Swift, RxSwift, SnapKit, Alamofire, Swinjext, Realm, Kotlin, RxKotlin&RxJava2 RxAndroid, Kodein, Google Databinding**

Holiday Business

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Windows UWP, iOS, macOS and Android applications
AUGUST 2014 – NOVEMBER 2019

A 2D turn-based strategy game with 3D elements, ideally similar to the Monopoly game, designed for computers and mobile platforms. Game kernel written in C++ with approximately 20,000 lines of code. Game functionalities: single and local network gameplay, saving the game state/options in the device memory, in-app purchases. Features of the game (visualization of the board in 3D, purchase/sale of land, houses and guesthouses, transaction and auction systems, etc.). For iOS, macOS and Android platforms, the main game core is based on cocos2d-x, while for Windows it is based on DirectX 11.1 in the style of Windows Universal App. Selected options storing, network communication and in-app purchases written in native technologies to the platform.
Used technologies: **C++, Swift, Kotlin, cocos2d-x, CoreData, Realm, DirectXTK, FW1FontWrapper**

Windows *iOS* *Android*

ADDITIONAL SKILLS

Foreign languages:

English – **B2+/C1** level - knowledge of technical programming vocabulary

Good knowledge:

Python, Keras + Tensorflow, scikit-learn, SciPy, NumPy, matplotlib, Matlab, Swift, Objective-C, Kotlin, iOS SDK(CoreData, AutoLayout, GCD itp.), Git, Java SE, Android SDK, Android frameworks(Maven/Gradle), RFO(RxSwift, RxKotlin itp.), Cobol, DB/2 IBM SQL, Windows Universal Apps SDK, C/C++, PHP, HTML, WinApi, MySQL, MS Visual Studio, Xcode, Android Studio

Basic knowledge:

JavaScript, jQuery, WinForms, C# & .NET, programów: Autodesk 3ds max, Inkscape

Interests:

- software development, especially creating end-user functional applications
- tracking the development of road infrastructure
- cycling
- good movie
- ice skating